

Week 2 Deep Learning Technical Report

1. Objective

The objective of Week 2 was to build and evaluate deep learning models using PyTorch. The work included implementing a neural network workflow, training a convolutional neural network (CNN) for image classification, and conducting controlled experiments with different optimizers and learning rates.

2. CNN Image Classification

A CNN was implemented in PyTorch for the CIFAR-10 image classification dataset. The dataset contained 50,000 training images and 10,000 testing images. The CNN used convolutional layers, ReLU activations, max-pooling layers, and fully connected layers. The model was trained for 5 epochs using CrossEntropyLoss and Adam optimization.

During the initial training run, the average training loss decreased from 1.3469 in Epoch 1 to 0.6218 in Epoch 5, indicating that the model was learning from the training data. An initial test accuracy of 70.93% was obtained.

3. Hyperparameter Experimentation

Controlled experiments were performed by changing the optimizer and learning rate while keeping the CNN architecture and number of epochs fixed at 5. The purpose was to compare training configurations and identify the best-performing setup.

Experiment	Optimizer	Learning Rate	Epochs	Test Accuracy
Baseline	Adam	0.001	5	72.70%
Adam – lower LR	Adam	0.0005	5	70.56%
Adam – high LR	Adam	0.01	5	10.00%
SGD	SGD	0.001	5	30.82%

4. Results and Findings

The best result was achieved by Adam with a learning rate of 0.001, reaching 72.70% test accuracy. Reducing the learning rate to 0.0005 resulted in a slightly lower accuracy of 70.56%. Increasing the learning rate to 0.01 caused a major drop in performance to 10.00%, showing that an overly large learning rate was unsuitable for this experiment. Replacing Adam with SGD at a learning rate of 0.001 also produced lower performance, with 30.82% test accuracy.

The training-loss results from the initial CNN run also showed a consistent decrease across the five epochs, from 1.3469 to 0.6218. Overall, the experiments indicate that the optimizer and learning rate had a strong effect on the model's performance, and Adam with a learning rate of 0.001 was the best configuration among the tested settings.

5. Conclusion

Week 2 successfully covered CNN-based image classification and hyperparameter experimentation in PyTorch. The final experiments provided a practical comparison of training configurations and identified Adam with a learning rate of 0.001 as the best-performing setup, with 72.70% test accuracy on CIFAR-10.